

CHEM&121: Introduction to Chemistry

Fall 2026 (September 21 – December 10, 2026)

What's Covered

Everything covered in this course, organized by topic and module — useful for comparing against an equivalent course at another institution, and for previewing what's ahead before the term starts.

Activity numbers below are provisional (carried over from the prior workbook) pending the Fall 2026 Chemfolio update — topics, labs, and objectives are current.

Module 1: Math and Measurement, Introduction to the Atom

What this module covers:

- States of matter; classifying matter as pure substance, mixture, element, or compound
- Metric measurement of length, mass, and volume
- Significant figures and scientific notation
- Unit conversion and temperature conversion
- Density and specific gravity, including mass/volume calculations
- Lab: Lab Safety Orientation and Scavenger Hunt
- Required Reading: Smith Ch. 1 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA2, CA3, CA4, CA5, CA6
- OOL1 Discussion
- ALEKS Problem Set: Ch. 1
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to drop online / 100% refund

Module 2: Density, Atomic Number, Ionization Energy

What this module covers:

- Identifying elements by symbol; classifying as metal, nonmetal, or metalloid
- Basic parts of the atom; isotopes
- Basic features of the periodic table (PTOE)
- Electronic structure of the atom; valence electrons
- Using the PTOE to predict relative atomic size and ionization energy
- Lab: Density and Sea Water
- Required Reading: Smith Ch. 2 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA7, CA9, CA10A, CA10B, Periodic properties (handout)
- OOL2 Discussion
- ALEKS Problem Set: Ch. 2
- CLOs addressed: 1 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to add/drop without a "W"

Module 3: Covalent Molecules, Shapes, Polarity and IMF

What this module covers:

- Predicting molecular shape; polar vs. nonpolar bonds and molecules
- Energy and units of energy; states of matter
- Intermolecular forces (IMF) and their effect on boiling/melting point
- Energy changes during a change of state; heating/cooling curves
- Lab: Molecular Shapes and GHGs
- Required Reading: Smith Ch. 3 (3.7–3.12), Ch. 4 (all, skip 4.5)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA13, CA16, CA17, CA27
- OOL3 Discussion
- ALEKS Problem Set: Ch. 4
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to withdraw for the quarter
- Also this module: Quiz 1 (Modules 1 and 2)

Module 4: Electrons, Bonding, and Naming Compounds

What this module covers:

- Electron configuration (Periods 1–3); electron-dot symbols
- Ionic vs. covalent bonds; the octet rule
- Writing and naming ionic compounds, including common polyatomic ions
- Lewis structures and naming for covalent compounds
- Lab: Copper Cycle and AMD
- Required Reading: Smith Ch. 3 (3.1–3.6)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA11, CA12, CA15, CA19
- OOL4 Discussion
- ALEKS Problem Set: Ch. 3
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day for 50% refund

Module 5: The Mole and Stoichiometry

What this module covers:

- Writing and balancing chemical equations
- Classifying reactions (combination, decomposition, single/double replacement)
- Oxidation, reduction, and redox reactions
- The mole, Avogadro's number, and molar mass
- Mole/mass calculations in chemical equations
- Lab: Soap and Soil Quality (Part 1)
- Required Reading: Smith Ch. 5 (5.1–5.8)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA18, CA20A, CA20B, CA22
- OOL5 Discussion
- ALEKS Problem Set: Ch. 5
- CLOs addressed: 2 (full text on the Course Learning Outcomes page)
- Also this module: Quiz 2 (Modules 3 and 4)

Module 6: Chemical Reactions: Energy, Rates, and Equilibrium

What this module covers:

- Endothermic vs. exothermic reactions; energy changes in reactions
- Factors affecting reaction rate
- Chemical equilibrium and Le Châtelier's principle
- How temperature is regulated in the body
- Lab: Soil Quality (Part 2)
- Required Reading: Smith Ch. 5 (5.9–5.12)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: Energy in Chemical Reactions Parts 1–3
- OOL6 Discussion
- ALEKS Problem Set: Ch. 5
- CLOs addressed: 3 (full text on the Course Learning Outcomes page)

Module 7: Gases

What this module covers:

- Measuring pressure and converting between pressure units
- Relationship between pressure, volume, and temperature (gas laws)
- Relationship between gas volume and moles; the ideal gas law
- Dalton's law and partial pressures
- Ozone and carbon dioxide as atmospheric components

- Lab: Air Pollution and Pollinators
- Required Reading: Smith Ch. 6 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: EOGL
- OOL7 Discussion
- ALEKS Problem Set: Ch. 6
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)

Module 8: Solutions, Concentration, Osmosis, and Acids

What this module covers:

- Fundamental properties of solutions; solution vs. colloid vs. suspension
- Electrolytes vs. nonelectrolytes
- Solubility prediction; effects of temperature and pressure on solubility
- Solution concentration calculations; preparing dilutions
- Osmosis and its relationship to biological membranes and dialysis
- Lab: Drinking Water
- Required Reading: Smith Ch. 7 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA28, CA29A, CA14, CA30A
- OOL8 Discussion
- ALEKS Problem Set: Ch. 7
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)
- Also this module: Quiz 3 (Modules 5 and 6)

Module 9: Acids, Bases, and Buffers

What this module covers:

- Identifying acids and bases; their characteristics
- Writing acid-base reaction equations; strong vs. weak acids/bases
- The ion product of water; calculating hydronium/hydroxide concentration
- Calculating pH; drawing products of common acid-base reactions
- Buffers and their role in maintaining pH in the body
- Lab: Surface and Ground Water
- Required Reading: Smith Ch. 8 (8.1–8.6, 8.8–8.9)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA30B, CA30C, CA31A, CA31B
- OOL9 Discussion
- ALEKS Problem Set: Ch. 8
- CLOs addressed: 1, 2, 3 (full text on the Course Learning Outcomes page)

- Also this module: Quiz 4 (Modules 7 and 8)

Module 10: Nuclear Chemistry

What this module covers:

- Types of radiation emitted by a radioactive nucleus
- Writing nuclear reaction equations; half-life
- Units used for measuring radioactivity
- Common radioisotopes used in medical diagnosis and treatment
- Nuclear fission and fusion; medical imaging without radioactivity
- Lab: Radiation and Alginate Beads
- Required Reading: Smith Ch. 9 (all)
- Final Project: Teach Your Community (TYC) due

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA8
- OOL10 Discussion
- ALEKS: Module 10
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)